

Activity 3.1.5 Kirchhoff's Laws Lab

PLTW

KIRCHHOFF'S LAWS LAB



Follow the Rules!



Branches and Nodes



Kirchhoff's Current Law



Kirchhoff's Voltage Law



Conclusion

ACTIVITY SUPPORTS



Sources

Follow the Rules!

GOALS	MATERIALS	RESOURCES
Experimentally validate Kirchhoff's laws.		
GOALS	MATERIALS	RESOURCES
- Variable power supply - Breadboard 100 Ω resistors 150 Ω resistors 220 Ω resistors 330 Ω resistors 470 Ω resistors		

- Hookup wires
- Digital multimeter

GOALS	MATERIALS	RESOURCES
• <u>Kirchhoff's Laws (Microsoft Excel)</u> • <u>Kirchhoff's Laws (Google Sheets)</u> • <u>Activity 3.1.5 Kirchhoff's Rule (Downloadable PDF)</u>		

When designing electrical circuits, engineers follow simple mathematical rules. Some of these rules, like those that govern the current and voltage in series and parallel circuits, might seem second nature to you by now. But what about when circuits become more complicated? That's where Kirchhoff's laws come into play!

Discovered by and named after the 19th-century German physicist Gustav Kirchhoff, these laws are like the secret codes that unlock the mysteries of electricity. They are the rules that help us understand how current and voltage behave in complex circuits.

In this activity, you'll walk in the footsteps of Gustav Kirchhoff and design, test, and analyze a complex circuit to develop rules that govern current and voltage. So, are you ready to embark on this electrifying journey of discovery?

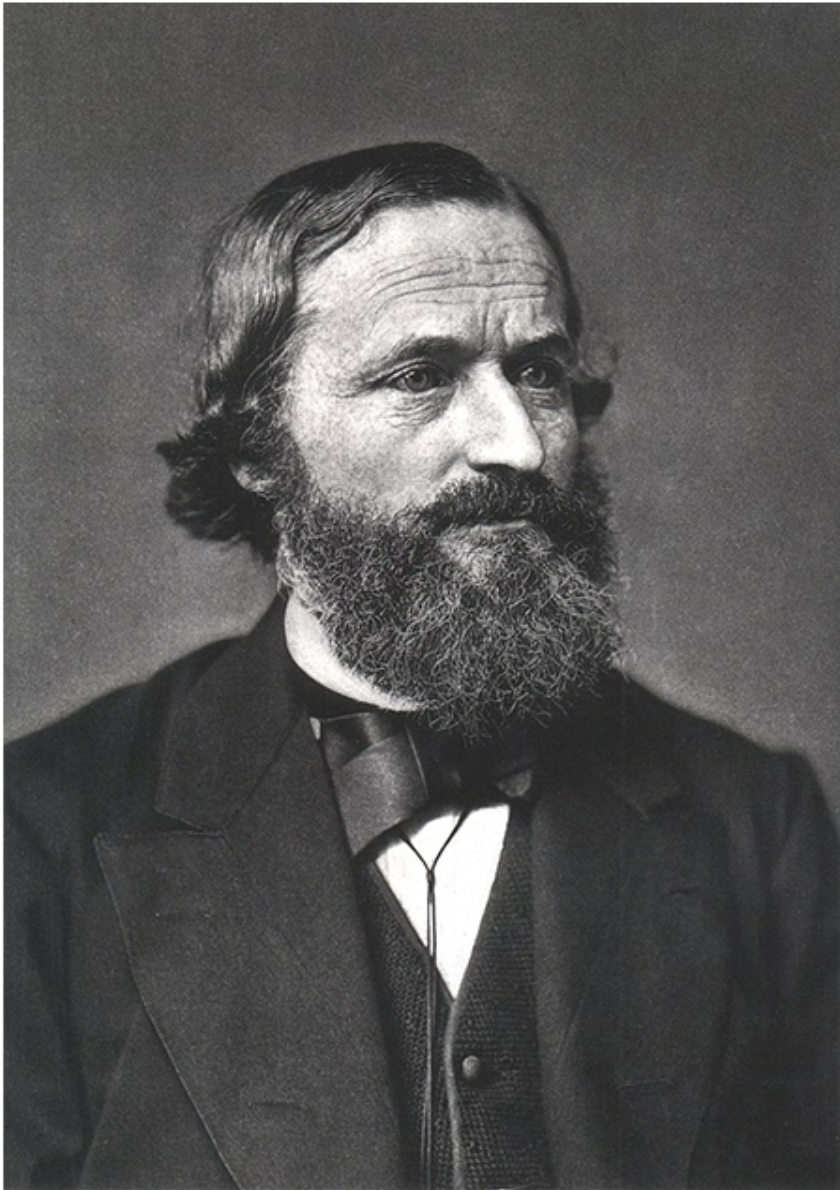


Figure 1. Gustav Kirchhoff¹

Branches and Nodes

Figure 2 shows a simple circuit composed of a power supply and a resistor.

Two important circuit elements are labeled on the diagram.

- The first is **branches**. The power supply is a branch because it has two terminals (positive and negative), which connect to the rest of the circuit through wires. The resistor is a branch because it has two leads that connect to the rest of the circuit through wires.
- The two branches meet at each of the two **nodes**. The positive end of the power supply is connected to one side of the resistor at a node. The other side of the resistor is connected to the negative side of the power supply at the other node.

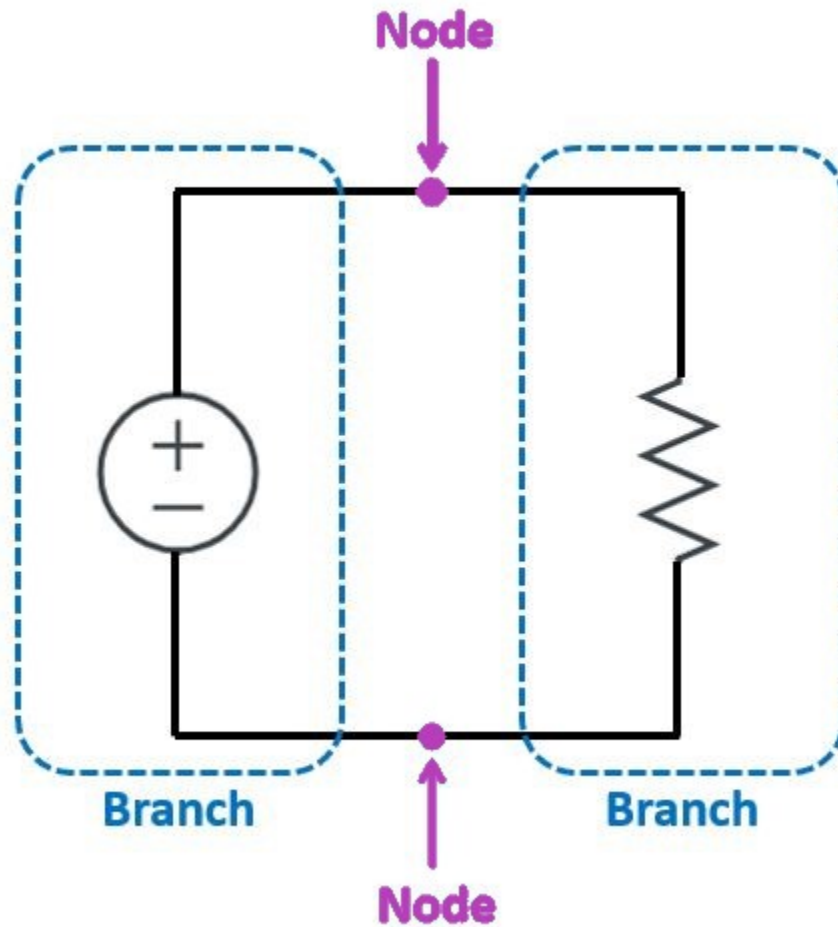


Figure 2. Branches and Nodes

Identifying branches and nodes is an important electrical engineering skill. Specifically, this helps engineers,

- Design well-organized and functional circuit layouts.
- Calculate current flow at various points within circuits.
- Calculate voltage drops across points within circuits.
- Systematically analyze and troubleshoot circuits to identify and rectify problems.

1

Observe the schematics of a series circuit with two resistors in Figures 3 and 4.

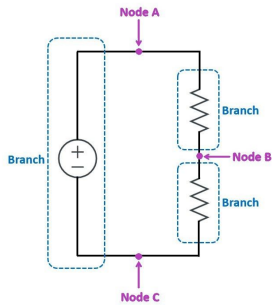


Figure 3. Series Circuit Schematics with Three Nodes

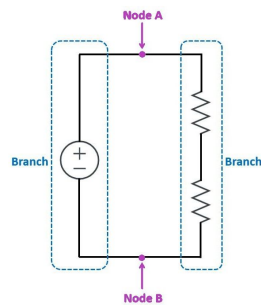


Figure 4. Series Circuit Schematics with Two Nodes

As shown in the figure, you can consider the pair of resistors as separate branches (Figure 3), which means the circuit has three nodes. Or you can consider the resistors as a single branch (Figure 4), in which case, the circuit has two nodes. Either grouping is acceptable.

2

Observe the schematic in Figure 5.

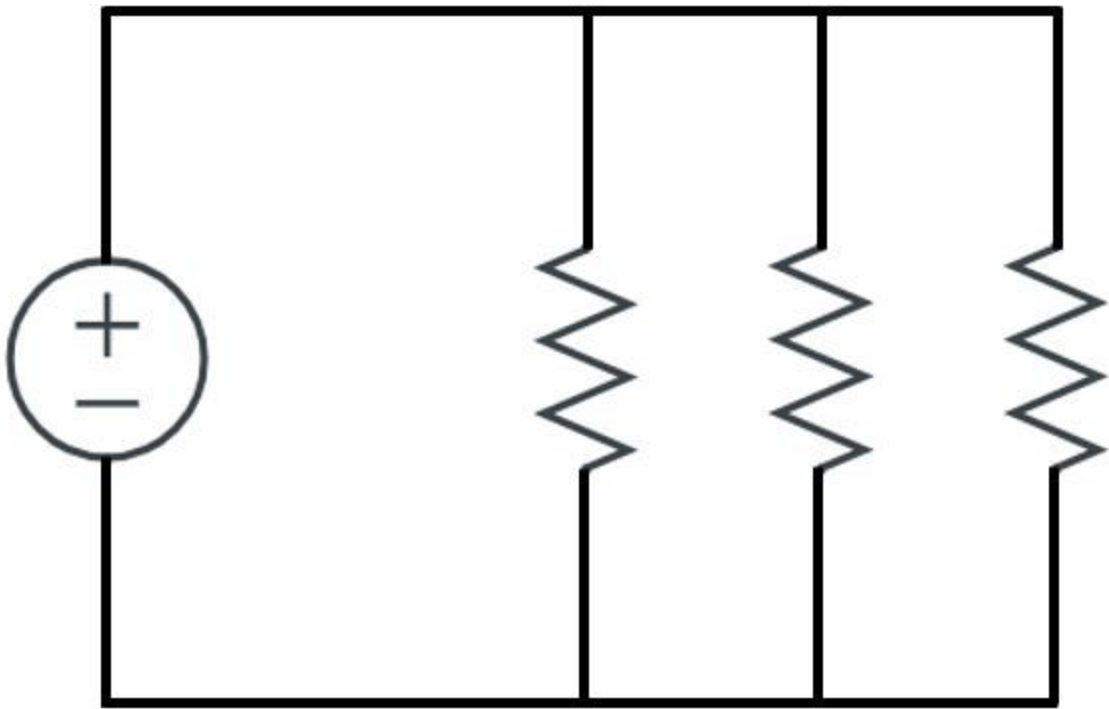


Figure 5. Parallel Circuit

Turn and Talk



How many nodes and branches are in the circuit in Figure 5?

HINT: Refer to the definitions.

Time to Practice

3

Identify the nodes in the circuit in Figure 6.

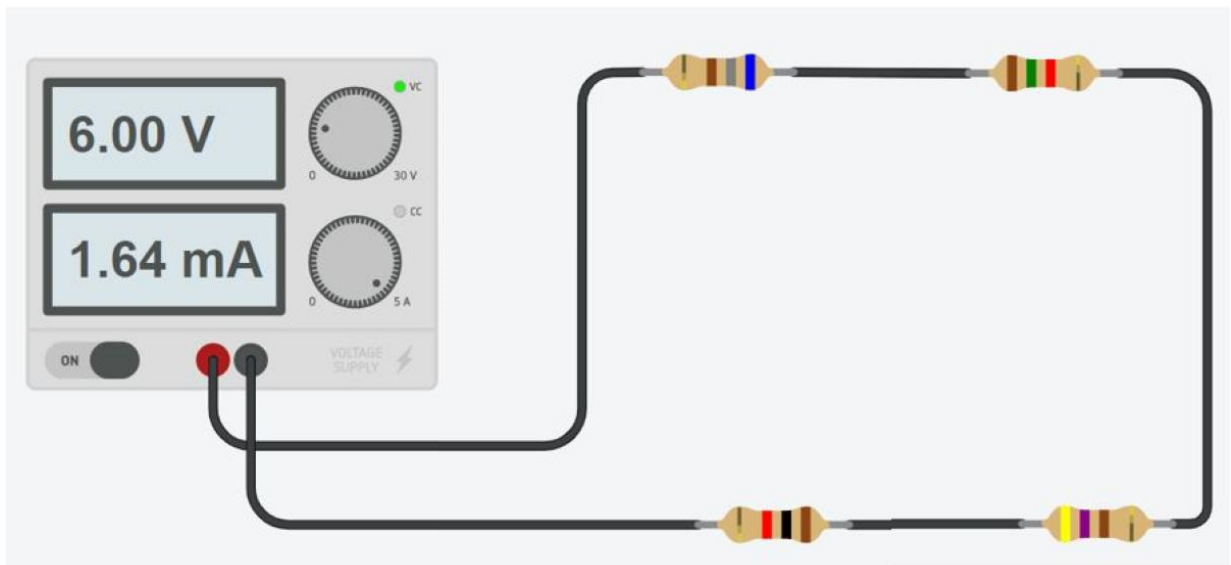


Figure 6. Resistors in Series

4

Identify the nodes in the circuit in Figure 7.

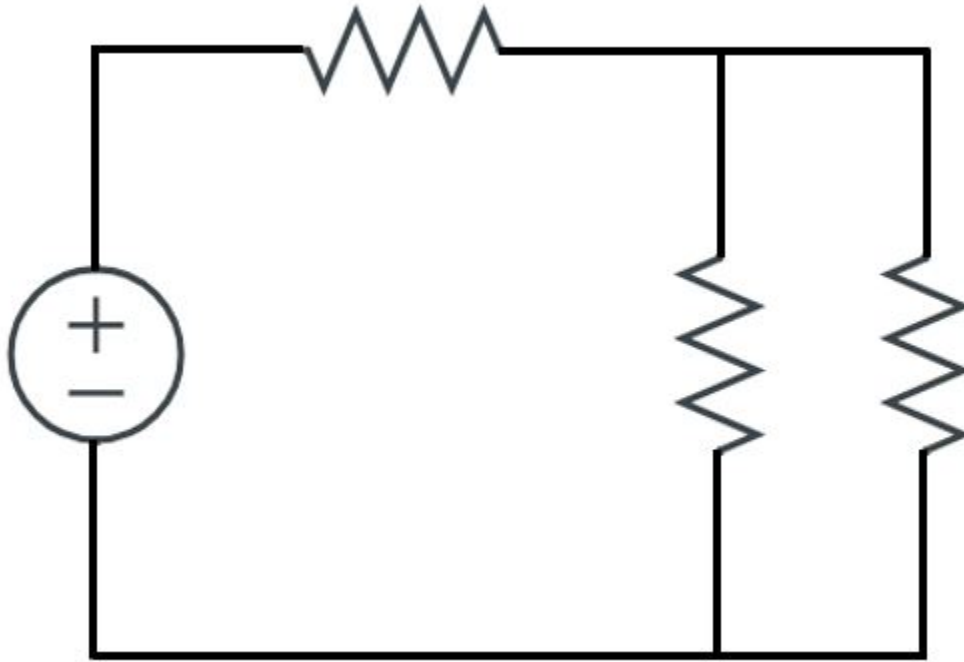


Figure 7. Combination Circuit

5

Draw a combination circuit with five nodes.

Kirchhoff's Current Law

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Using a breadboard and resistors, build the circuit in Figure 8. Your team can decide which resistors to use at each location, and what voltage to set the power supply. The nodes are indicated on the schematic.

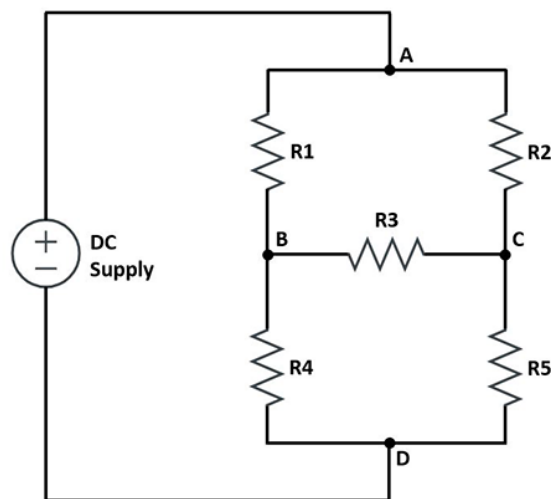


Figure 8. Schematic Diagram

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Download and save a copy of [Kirchhoff's Laws \(Microsoft Excel\)](#) or [Kirchhoff's Laws \(Google Sheets\)](#).

8

In the Resistance (Ω) column of the Kirchhoff's Laws spreadsheet, record the resistance values that your team selected. Disconnect the power supply and measure the Total Resistance of the circuit to record that value, as well.

9

Use the multimeter to measure the voltage across each resistor. Record the values in the Voltage (V) column of the spreadsheet.

10

Use Ohm's Law to calculate the current through each resistor. Record the values in the Current (A) column of the spreadsheet.

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Quantitatively compare the current entering node A to the current exiting node A. Figure 9 shows current entering and exiting node A.

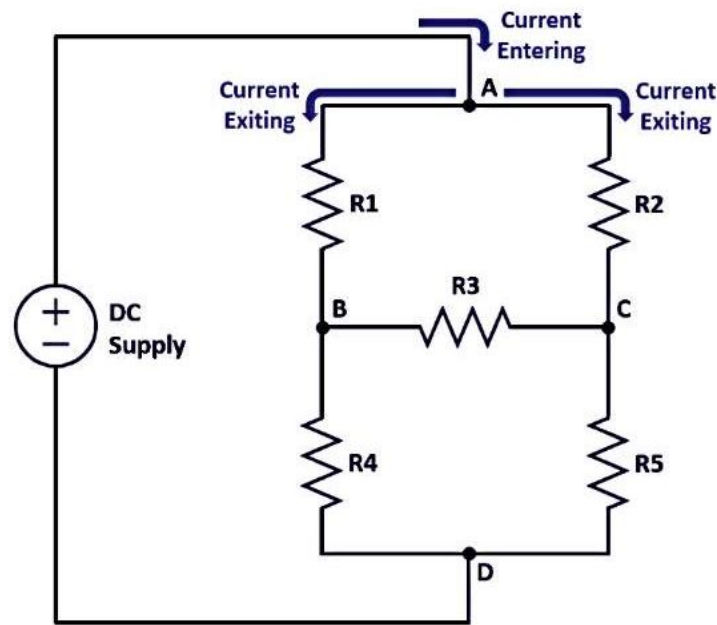


Figure 9. Node A Current

- The current entering node A is equal to the total circuit current.
- The current exiting node A is the current through resistor 1 plus the current through resistor 2.

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In the spreadsheet, record these values under the Current In and Current Out columns for Node A.

Quantitatively compare the current entering and exiting each of the other three nodes. Pay special attention to the current through R_3 . Depending upon the resistor values you selected, the current could be flowing left or right through the resistor (positive or negative in your quantitative comparison). Record all values on the spreadsheet.

Reflection



- What do you notice about the net current for each node of your circuit?
- Why do you think this is true?
- How would you write this as a mathematical statement?

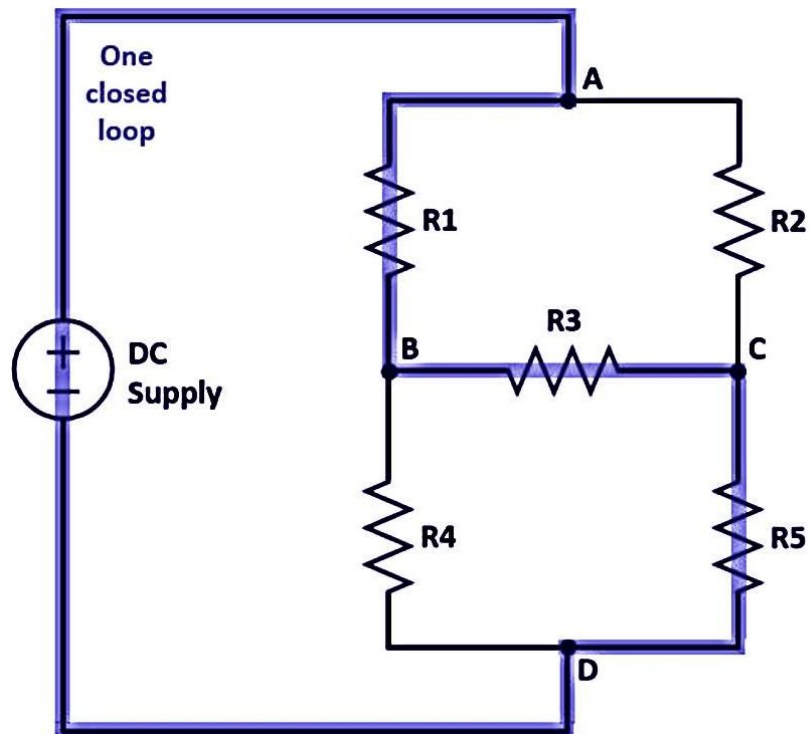
Kirchhoff's Voltage Law

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Observe the circuit you made using Figure 8. How many closed loops (circuits) are there through which current can flow?

Example Closed Loop

This example closed loop goes through the power supply, R1, R3, and R5.



Hint 1 —

Did you find four? If so, nice work. But there are three more!

Hint 2 —

A closed loop does not have to go through the power supply of the circuit. It can be any closed loop somewhere on the circuit.

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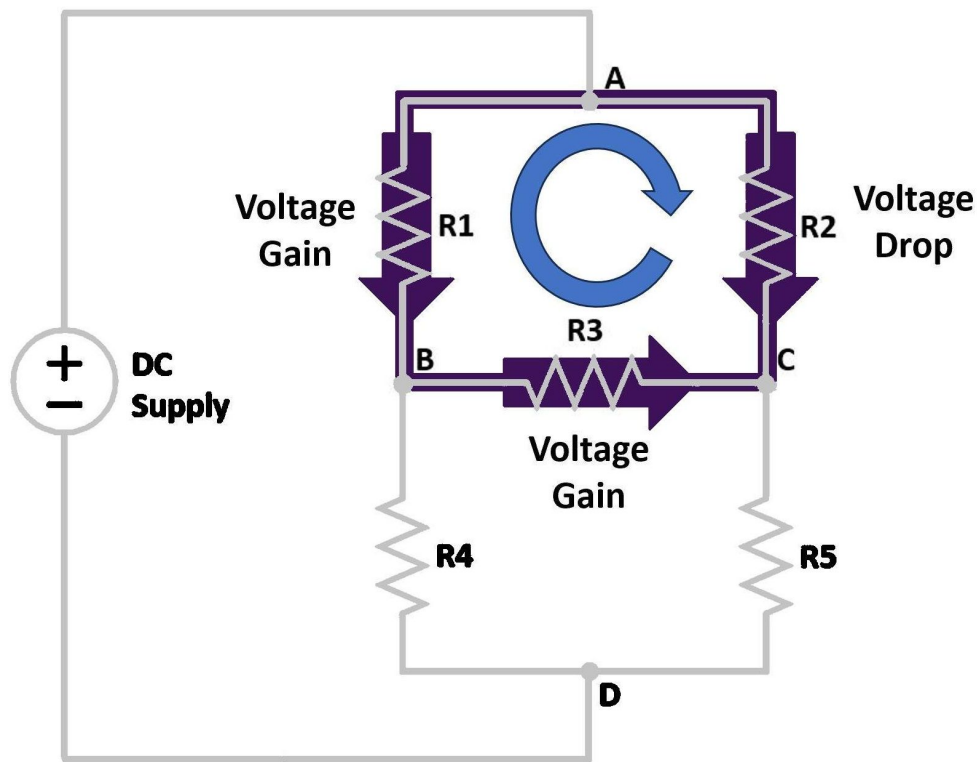
Quantitatively compare the voltage drops to the voltage gains for each loop.

- Pick any point on a loop as a starting point and travel clockwise (arbitrary), summing the voltage drops and gains along the way.
 - When you travel in the same direction as current flow and encounter a resistor, there is a voltage drop.
 - When you travel in the opposite direction from current flow and encounter a resistor, there is a voltage gain.
 - When you travel through a power supply from negative to positive, there is a voltage gain.
 - When you travel through a power supply from positive to negative, there is a voltage drop.

HINT: Start on one of the smaller loops (one that does not go through the power supply).

Example Loop Calculation —

- Starting at node A and traveling clockwise, you first encounter R2, in the direction the current is traveling. This results in a voltage drop of 1.44 V, which you can indicate as -1.44 V .
- Continuing clockwise, you encounter R3 opposite the direction the current is traveling. This results in a voltage gain of 0.03 V, which you can indicate as 0.03 V.
- Continuing clockwise, you encounter R1 opposite the direction the current is traveling. This results in a voltage gain of 1.41 V, which you can indicate as 1.41 V.
- Voltage gain(s) = $1.41\text{ V} + 0.03\text{ V} = 1.44\text{ V}$
- Voltage drop(s) = 1.44 V
- Net voltage = $1.44\text{ V} + -1.44\text{ V} = 0\text{ V}$



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In the spreadsheet, record these values under the Voltage Drops and Voltage Gains columns for Closed Loops i–vii.

17

Quantitatively compare the voltage drops to the voltage gains (the net voltage) for each of the seven closed loops. Record all values on the spreadsheet.

Reflection



- What do you notice about the net voltage of each closed loop of your circuit?
- Why do you think this is true?
- How would you write this as a mathematical statement?

Conclusion

Question 1

Explain how you can use Kirchhoff's Current Law to verify that the current through each of the resistors in Figure 10 must be the same. Draw diagrams to support your explanation as necessary.

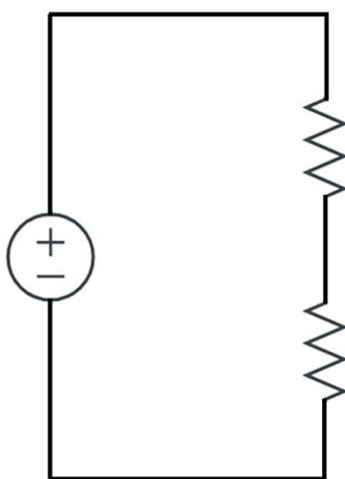


Figure 10. Series Circuit

Question 2

Explain how you can use Kirchhoff's Voltage Law to verify that the voltage across each of the resistors in Figure 11 must be the same. Draw diagrams to support your explanation as necessary.

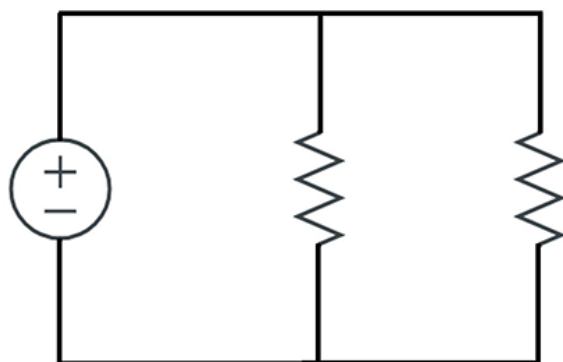


Figure 11. Parallel Circuit

Sources

Third-party images, videos, and Historical Highlight information used throughout this activity were drawn from the following source:

1. Portrait of Gustav Kirchhoff, Smithsonian Libraries, permitted under [CC0 Public Domain](#)